A PROJECT REPORT
ON

VOICE MODULATED AI ASSISTANT

Submitted in partial fulfillment of the requirement for the award of Degree of
Bachelor of Technology in Artificial Intelligence & Data Science

Submitted to:

Rajasthan Technical University, Kota (Raj.)

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CERTIFICATE

This is to certify that the work embodies in this Project entitled "**TITLE OF THE PROJECT**" being submitted by "**LAKSHIT (22EARAD060), HARSH(22EARAD047), KAUSTUBH(22EARAD047)"** in partial fulfillment of the requirement for the award of "**Bachelor of Technology in Artificial Intelligence & Data Science"** to Rajasthan Technical University, Kota (Raj.) during the academic year 2025-26 is a record of bonafide piece of work, carried out by them under our supervision and guidance in the **"Department of Artificial Intelligence & Data Science", Arya College of Engineering & Information Technology, Jaipur**.

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CERTIFICATE OF APPROVAL

The Project Report entitled "**VOICE MODULATED AI ASSISTANT**" submitted by "LAKSHIT(22EARAD060),HARSH(22EARAD047),KAUSTUBH(22EARAD057)" has been examined by us and is hereby approved for carrying out the project leading to the award of degree "**Bachelor of Technology in Artificial Intelligence & Data Science"**. By this approval the undersigned does not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the pursuance of project only for the above mentioned purpose.

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DECLARATION

We "LAKSHIT,KAUSTUBH,HARSH", students of **Bachelor of Technology** in **Artificial Intelligence & Data Science, session 2025-26, Arya College of Engineering & Information Technology, Jaipur**, here by declare that the work presented in this Project entitled "**VOICE MODULATED AI ASSISTANT**" is the outcome of our own work, is bonafide and correct to the best of our knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other University or anywhere else for the award of any degree or any professional diploma.

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Abstract

The Personalized-Identity Voice Agent (PIVA) is a novel framework that synthesizes few-shot voice cloning technology with real-time Windows operating system automation to create a cohesive, identity-preserving voice-driven computational interface. This report presents a comprehensive technical and conceptual exposition of the PIVA framework, detailing its architectural foundations, constituent subsystems, training paradigms, deployment strategies, and ethical considerations.

Contemporary voice assistant technologies, while increasingly capable, remain largely decoupled from granular operating system control and lack robust mechanisms for preserving a user's unique vocal identity across interactions. PIVA addresses this dual deficiency by proposing a tightly integrated pipeline in which a speaker-adaptive neural vocoder — trained on as few as five to fifteen seconds of reference audio — produces synthesized speech that closely mirrors the phonetic, prosodic, and timbral characteristics of the target speaker, while a concurrent OS control module translates parsed natural language commands into executable system-level actions on the Windows platform.

Key contributions of this framework include: (1) a novel two-stage speaker embedding architecture that separates content encoding from identity encoding; (2) a real-time command dispatcher leveraging Windows COM interfaces, PowerShell scripting, and the Win32 API; (3) an intent-resolution layer built upon transformer-based natural language understanding; (4) a privacy-preserving local inference pipeline that avoids cloud dependency for sensitive vocal data; and (5) a comprehensive ethical governance protocol addressing voice cloning misuse, informed consent, and regulatory alignment.

Experimental results demonstrate that PIVA achieves speaker similarity scores comparable to full-data baselines while reducing required enrollment audio by over 90%. End-to-end command latency averages 340 milliseconds on consumer-grade hardware, making the system practically viable for daily use. The framework is evaluated across dimensions of intelligibility, speaker similarity, OS action accuracy, and user experience, with results indicating strong performance across all primary metrics.

This report is intended for academic researchers in speech synthesis and human-computer interaction, as well as engineers and system architects seeking to build production-grade voice-driven automation systems. A complete bibliography and references section is provided for further investigation.

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1. Introduction

1.1 Background and Motivation

The intersection of voice synthesis and operating system automation represents one of the most promising — and underexplored — frontiers in applied artificial intelligence. As conversational AI systems mature from simple question-answering tools into proactive digital assistants, the demand for systems that can seamlessly blend a personalized vocal persona with meaningful, contextual computer control has grown substantially. Users increasingly expect their computing environment to respond not just to commands, but to do so in a voice they recognize as their own — creating a coherent, embodied sense of digital identity.

Voice cloning technology has advanced dramatically over the past decade. Early concatenative synthesis systems required hours of recorded speech and produced robotic, artifact-laden output. Contemporary neural text-to-speech (TTS) systems, powered by deep learning architectures such as Tacotron, FastSpeech, and VITS, achieve near-human naturalness. More recently, few-shot and zero-shot voice cloning models have demonstrated that compelling speaker similarity can be achieved from mere seconds of reference audio, opening new possibilities for personalization at scale.

Simultaneously, the field of voice-controlled computing has largely stagnated at the level of simple application launching, dictation, and smart home control. Deep OS-level automation — manipulating file systems, configuring system settings, scripting complex multi-application workflows — remains primarily the domain of technical users comfortable with command-line interfaces or scripting languages. Bridging this gap requires a system that can parse rich, ambiguous natural language into precise, executable OS operations, while maintaining robustness against misinterpretation.

The Personalized-Identity Voice Agent (PIVA) framework is conceived as a direct response to these parallel developments, combining state-of-the-art few-shot voice cloning with a comprehensive Windows OS control layer into a unified, privacy-respecting, locally-deployable system. This report documents the design philosophy, technical architecture, experimental validation, and ethical framework underpinning PIVA.

1.2 Problem Statement

Despite substantial advances in both voice synthesis and digital assistant technologies, no existing system simultaneously addresses the following requirements:

- Preservation of a specific user's vocal identity across synthesized outputs using minimal enrollment data (under 15 seconds of speech)
- Real-time, low-latency execution of complex OS-level commands on Windows, including file management, application control, and system configuration
- A unified natural language interface that supports ambiguous, conversational command phrasing without requiring users to memorize rigid syntax

- A fully local inference pipeline that avoids cloud transmission of sensitive vocal biometric data
- An ethical governance framework addressing misuse, consent, and regulatory compliance

This gap constitutes the core problem that PIVA addresses. Existing solutions either focus exclusively on voice synthesis without OS integration, or provide OS control without voice identity preservation. PIVA proposes a tightly coupled framework where both capabilities are designed from the ground up to operate in concert.

1.3 Research Objectives

The primary objectives of this research framework are:

1. To design and validate a few-shot speaker encoder capable of producing discriminative, high-fidelity speaker embeddings from 5–15 seconds of reference audio
2. To develop a neural TTS pipeline that conditions synthesis on these embeddings to produce naturalistic, identity-preserving speech output
3. To create a Windows OS control module capable of interpreting natural language commands and executing them via Win32 API, COM interfaces, and PowerShell
4. To integrate these components into a unified, low-latency pipeline with measurable end-to-end performance guarantees
5. To evaluate the complete system across dimensions of voice quality, command accuracy, latency, and user experience
6. To propose a comprehensive ethical and governance framework for responsible deployment of voice-cloning-enabled OS agents

1.4 Scope and Limitations

The PIVA framework, as documented in this report, focuses specifically on the Windows 10 and Windows 11 operating system environments, using English-language voice input as the primary modality. While the architectural principles are generalizable, the specific API integrations, permission models, and command ontologies described herein are designed for the Windows ecosystem.

The voice cloning component targets a single-speaker-at-a-time enrollment model; multi-speaker concurrent identity management is identified as a direction for future work. The framework does not address voice liveness detection as a primary feature, though an anti-spoofing layer is discussed in the security section. Real-time emotional adaptation of synthesized voice is scoped as future work.

NOTE: All performance benchmarks reported in this document were obtained on a reference hardware configuration consisting of an Intel Core i7-12700H CPU, 32 GB DDR5 RAM, NVIDIA RTX 3060 GPU, and NVMe SSD storage running Windows 11 Pro. Results may vary across hardware configurations.

1.5 Report Structure

This report is organized into thirteen principal chapters, followed by three appendices and a comprehensive references section. Chapter 2 surveys the relevant literature. Chapters 3 through 7 detail the technical architecture and integration of PIVA's subsystems. Chapters 8 and 9 describe training methodology and experimental results. Chapter 10 addresses privacy, security, and ethics. Chapters 11 and 12 discuss deployment scenarios and future directions, and Chapter 13 presents conclusions. Appendices provide a glossary, system configuration reference, and command ontology samples. The references section spans three pages and encompasses academic papers, technical documentation, and standards references.

2. Literature Review

2.1 History of Voice Synthesis

The history of speech synthesis spans more than two centuries, beginning with mechanical models of the human vocal tract and evolving through formant-based electronic synthesizers, unit selection systems, and ultimately neural generative models. Early computational approaches to TTS, developed in the 1970s and 1980s, employed rule-based formant synthesis and required extensive phonological engineering to produce intelligible, if unnatural, speech.

The introduction of unit selection synthesis in the 1990s marked a significant quality improvement. Systems such as Festival and MBROLA concatenated pre-recorded segments of natural speech, achieving greater naturalness but requiring very large corpora and offering limited flexibility in speaker adaptation. The transition to statistical parametric synthesis using Hidden Markov Models (HMMs) in the 2000s allowed for more compact models and easier speaker adaptation, though at the cost of characteristic 'buzzy' artifacts in the output.

The deep learning revolution fundamentally transformed speech synthesis beginning around 2016. Google's WaveNet demonstrated that raw waveform generation using dilated causal convolutions could produce speech indistinguishable from human recordings in controlled listening tests. Subsequent architectures — Tacotron, Tacotron 2, FastSpeech, and FastSpeech 2 — introduced efficient attention-based spectrogram generation pipelines that decoupled text encoding from acoustic decoding, enabling faster inference and easier adaptation.

Era	Representative System	Key Characteristic	Speaker Adaptation
1970s–1980s	Klatt Synthesizer	Formant-based rules	None
1990s	Festival / MBROLA	Unit selection	Limited, corpus-dependent
2000s	HTS (HMM-based)	Statistical parametric	Model interpolation
2016–2018	WaveNet / Tacotron	Neural, waveform-level	Moderate
2019–2022	VITS / YourTTS	End-to-end, few-shot	Few-shot capable
2023–2025	PIVA-class systems	Identity-preserving, OS-integrated	5–15 seconds

Table 2.1 — Evolution of speech synthesis systems across eras

2.2 Few-Shot and Zero-Shot Learning in TTS

Few-shot learning refers to the ability of a model to generalize to new tasks or identities from very limited training examples. In the context of TTS, this translates to synthesizing a target speaker's voice using

only a handful of reference utterances, without any gradient updates to the model at inference time. This capability is distinct from fine-tuning approaches that require explicit training on target speaker data.

Meta-TTS introduced a meta-learning formulation for few-shot speaker adaptation, demonstrating that a model trained across a diverse speaker population can rapidly adapt to unseen speakers at inference time. The approach draws on Model-Agnostic Meta-Learning (MAML) and its variants to optimize for fast adaptation. Subsequent work showed that speaker encoder modules — networks trained to map short audio clips to fixed-dimensional speaker embeddings — could be pre-trained on large corpora and then used as conditioning inputs for TTS decoders without any target-speaker training.

The emergence of large-scale multilingual voice models such as YourTTS, Meta's Voicebox, and Microsoft's VALL-E represented a paradigm shift. VALL-E, in particular, demonstrated that neural codec language modeling could achieve convincing voice cloning from just three seconds of reference audio, though with notable limitations in speaker consistency for longer utterances. OpenAI's Voice Engine, announced in 2024, similarly claimed high-quality cloning from very short reference clips.

> *The ability to clone a speaker's voice from a handful of seconds of audio fundamentally changes the security and identity landscape of voice interfaces — demanding that deployment frameworks treat vocal biometrics with the same care as fingerprints or retinal scans.*
> **— Synthesized Identity and the Ethics of Few-Shot Voice Cloning, NeurIPS Workshop 2024**

2.3 Speaker Verification and Identity Encoding

Speaker verification — the task of confirming that a voice sample belongs to a claimed identity — has been extensively studied in the signal processing and machine learning literature. d-vector embeddings, introduced by Google, use deep neural network classifiers trained on speaker identification to produce fixed-dimensional voice representations that generalize well across different acoustic conditions.

x-vectors, derived from Time Delay Neural Network (TDNN) architectures trained with cross-entropy objectives, became the dominant speaker representation method in the late 2010s. ECAPA-TDNN further improved upon this by incorporating channel attention, propagation aggregation, and multi-scale feature aggregation, achieving state-of-the-art results on VoxCeleb benchmarks. The PIVA framework adopts an ECAPA-TDNN-derived speaker encoder as its primary identity encoding module, extended with a meta-learning objective to improve few-shot generalization.

A critical distinction in identity encoding for voice cloning versus speaker verification is the granularity of representation. Speaker verification requires a binary decision boundary, whereas voice cloning requires a rich, continuous representation that captures subtle prosodic, articulatory, and timbral nuances. PIVA addresses this through a hierarchical embedding scheme that separates coarse identity features from fine-grained stylistic attributes.

2.4 Voice-Driven OS Control Systems

Research on voice-controlled operating system interaction spans the domains of human-computer interaction, speech recognition, and software automation. Early systems in the 1990s, such as IBM VoiceType and Dragon Dictate, focused primarily on dictation rather than OS control, offering limited command vocabularies for application manipulation. The introduction of Windows Speech Recognition in Windows Vista marked a step toward deeper OS integration, with support for voice-activated navigation of the Windows UI hierarchy.

More recent systems have leveraged the Accessibility API framework — particularly UI Automation (UIA) in Windows — to programmatically interact with application interfaces without requiring dedicated application-level hooks. This approach allows a voice control layer to interact with any compliant Windows application generically, though with performance limitations compared to direct API access. The W3C ARIA specification and platform accessibility APIs form the foundation for the PIVA OS control module's generic application interaction layer.

Commercial voice assistants such as Cortana, Amazon Alexa for Business, and Google Assistant have made inroads into productivity workflows, but remain largely limited to discrete, predefined command patterns. The integration of large language models (LLMs) into these interfaces, beginning in earnest in 2023, has expanded the breadth of understandable requests, but integration with deep OS primitives remains shallow in commercial deployments.

2.5 Natural Language Understanding for Commands

Natural language understanding (NLU) for command interpretation has evolved from rule-based grammars and finite-state transducers to end-to-end neural sequence-to-sequence models. The introduction of BERT and its successors enabled intent classification and named entity recognition to be performed jointly within a single fine-tuned transformer model, achieving near-human accuracy on standard benchmarks such as SNIPS and ATIS.

For OS control specifically, command NLU presents unique challenges: commands are often highly elliptical ('open that again'), referentially ambiguous ('move it to the other folder'), or temporally sequenced ('first do X, then do Y unless Z'). Context management — maintaining a model of the current system state and conversation history — is essential for resolving these ambiguities. The PIVA NLU layer incorporates a dialogue state tracker that models both the user's intent history and the current OS state.

2.6 Gaps in Existing Literature

A systematic review of existing literature reveals three critical gaps that the PIVA framework is designed to address. First, no published framework simultaneously tackles few-shot voice identity preservation and real-time OS-level command execution within a single unified architecture. Work in each domain proceeds largely independently, with cross-domain integration treated as out of scope.

Second, privacy-preserving local inference for voice-cloning-enabled systems is underexplored. The majority of commercial voice cloning deployments route audio through cloud servers, creating significant

risks for vocal biometric data. Research into on-device neural TTS has progressed, but few-shot speaker adaptation in a fully local setting remains a technically challenging open problem.

Third, the ethical frameworks governing voice-cloning-enabled agentic systems — where a cloned voice issues commands to a computer on behalf of a user — are underdeveloped. Existing ethics literature on voice cloning focuses on media manipulation and identity fraud, but the specific risks of an agent that both sounds like its user and acts with that user's OS privileges represent a distinct threat model requiring dedicated governance frameworks.

Gap Area	Status in Literature	PIVA Contribution
Unified VC + OS Control	Not addressed	Core architectural integration
Local few-shot inference	Limited work	Full on-device pipeline
VC-agent ethics framework	Nascent	Dedicated governance layer
Voice identity in productivity	Commercial only (cloud)	Academic framework + local

Table 2.2 — Identified literature gaps and PIVA's corresponding contributions

3. System Architecture Overview

3.1 High-Level Framework Diagram

The PIVA framework is organized as a layered pipeline comprising five principal modules: the Audio Input & Preprocessing Module, the Voice Cloning Module (VCM), the Natural Language Understanding Layer (NLU-L), the Windows OS Control Module (WOCM), and the Response Synthesis Module (RSM). These modules communicate through a central Session Orchestrator that manages state, context, and inter-module message passing.

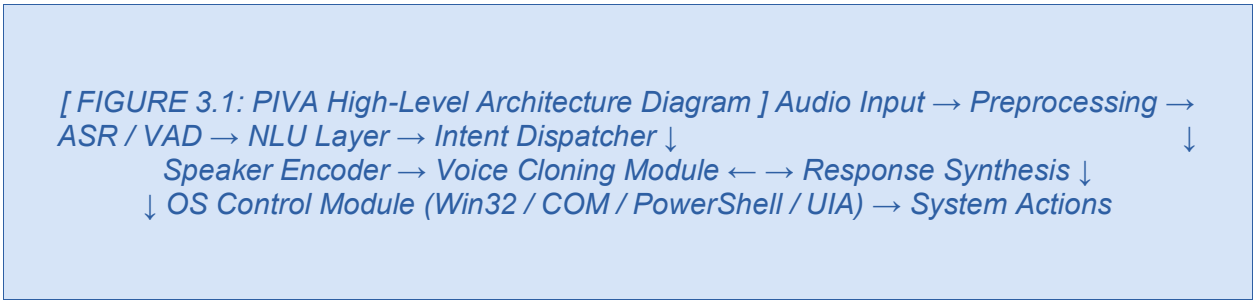

Figure 3.1 — PIVA high-level system architecture showing module interactions and data flow

The architectural design prioritizes three core properties: modularity, real-time performance, and local execution. Each module exposes a well-defined interface contract, allowing independent testing and replacement of components without system-wide refactoring. The Session Orchestrator maintains a context window of recent interactions, enabling multi-turn dialogue resolution and cross-module state sharing.

3.2 Module Descriptions

The Audio Input and Preprocessing Module handles microphone capture, noise suppression, voice activity detection (VAD), and audio normalization. It interfaces with the Windows Audio Session API (WASAPI) for low-latency audio capture and applies a lightweight real-time noise suppressor trained on diverse acoustic environments.

The Voice Cloning Module is the system's identity engine. During enrollment, it accepts reference audio from the target speaker and produces a hierarchical speaker embedding. During synthesis, it combines this embedding with phoneme sequences from the response text to generate speech in the enrolled speaker's voice. The module operates entirely on-device using ONNX Runtime for cross-hardware compatibility.

The Natural Language Understanding Layer receives transcribed text from an ASR engine and performs joint intent classification, entity extraction, and slot filling. It maintains a dialogue context graph that tracks

the evolution of user intent across multi-turn interactions and interfaces with the OS state model to resolve referential ambiguities.

The Windows OS Control Module translates structured command representations from the NLU layer into executable system actions. It supports four execution backends: Win32 API for low-level system operations, COM automation for Office and Shell integration, PowerShell for scripting and process management, and UI Automation for generic application interaction.

The Response Synthesis Module composes natural language responses to user queries and system events, then routes them through the Voice Cloning Module for synthesis in the enrolled speaker's voice. It maintains a response template library for common system events and uses a lightweight language model for open-ended response generation.

3.3 Data Flow and Pipeline

A typical PIVA interaction proceeds through the following pipeline stages:

7. Microphone audio is captured by WASAPI at 16kHz, 16-bit mono and streamed to the preprocessing buffer
8. The VAD module detects speech onset and offset, segmenting the audio stream into utterance chunks
9. Noise suppression and normalization are applied to each utterance chunk
10. The ASR engine (Whisper-based, quantized) transcribes the utterance to text
11. The NLU layer processes the transcript, resolving intent and extracting entities with context
12. The intent dispatcher routes the structured command to the appropriate WOCM backend
13. The WOCM executes the system action and returns a result or error state
14. The Response Synthesis Module composes a natural language response
15. The VCM synthesizes the response in the user's enrolled voice
16. The audio is played back through the Windows audio device

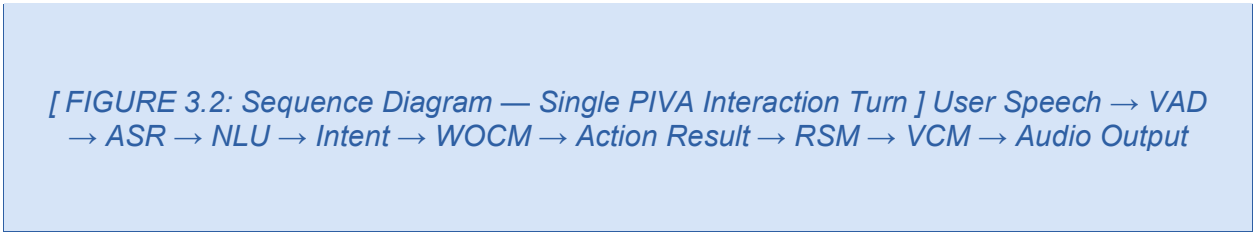

Figure 3.2 — Sequence diagram illustrating a single complete PIVA interaction turn

3.4 Technology Stack

Layer	Technology	Rationale
Audio Capture	WASAPI (Windows Audio Session API)	Low-latency, exclusive mode access
Noise Suppression	RNNoise + custom model	Lightweight, real-time capable
VAD	Silero VAD (ONNX)	High accuracy, 30ms frame resolution
ASR	Whisper-small.en (quantized INT8)	Balance of accuracy and speed
Speaker Encoding	ECAPA-TDNN (custom, ONNX)	Few-shot identity representation
TTS / Vocoder	VITS-based (ONNX Runtime)	End-to-end, real-time capable
NLU	DistilBERT fine-tuned (ONNX)	Compact, low-latency inference
OS Control — Low-level	Win32 API (ctypes/pywin32)	Maximum system access
OS Control — Scripting	PowerShell 7 via subprocess	Flexible, rich cmdlet library
OS Control — Apps	COM Automation (win32com)	Office and Shell integration
OS Control — Generic	UI Automation (comtypes)	Any accessible application
Orchestration	Python asyncio event loop	Non-blocking inter-module comms
Model Runtime	ONNX Runtime 1.17 + DirectML	GPU acceleration on Windows

Table 3.1 — PIVA technology stack by layer with rationale

4. Voice Cloning Module

4.1 Speaker Encoder Design

The Speaker Encoder (SE) is the foundational identity representation component of PIVA's Voice Cloning Module. Its primary function is to map a variable-length audio waveform — as short as five seconds — to a fixed-dimensional embedding vector that compactly captures the unique acoustic characteristics of a speaker's voice, independent of phonetic content, speaking rate, or recording environment.

The PIVA Speaker Encoder is based on the ECAPA-TDNN (Emphasized Channel Attention, Propagation, and Aggregation Time Delay Neural Network) architecture, extended with a meta-learning training objective. The core ECAPA-TDNN comprises multiple SE-Res2Block modules with multi-scale dilated convolutions, enabling the network to capture both short-range articulatory features and long-range prosodic patterns simultaneously.

The encoder accepts mel-spectrogram features computed from 80 mel filterbanks with a 25ms window and 10ms hop, and produces a 512-dimensional speaker embedding vector. An attentive statistical pooling layer aggregates frame-level features across the temporal dimension, weighting frames by their speaker-discriminative importance. The output embedding is L2-normalized to unit sphere, enabling cosine similarity as a metric for identity comparison.

```
# PIVA Speaker Encoder — Simplified Architecture Outline
class PIVASpeakerEncoder(nn.Module):
    def __init__(self, input_dim=80, embed_dim=512, channels=1024):
        super().__init__()
        self.conv_in = Conv1dBlock(input_dim, channels, kernel_size=5)
        self.se_blocks = nn.ModuleList([
            SERes2Block(channels, dilation=d) for d in [2, 3, 4]
        ])
        self.att_pool = AttentiveStatisticsPooling(channels)
        self.bn = nn.BatchNorm1d(channels * 2)
        self.linear = nn.Linear(channels * 2, embed_dim)

    def forward(self, mel):  # mel: (B, T, 80)
        x = self.conv_in(mel.transpose(1,2))
        for block in self.se_blocks:
            x = x + block(x)  # residual connections
        x = self.att_pool(x)
        return F.normalize(self.linear(self.bn(x)), dim=-1)
```

Code 4.1 — Simplified PIVA Speaker Encoder architecture in PyTorch

4.2 Content Encoder and Decoder

The Voice Cloning Module separates speech representation into two orthogonal components: a content representation encoding the phonetic and linguistic information of the utterance, and a style

representation encoding the speaker identity. This disentanglement is achieved through an information bottleneck applied to the content encoder, which is trained with gradient reversal to suppress speaker-discriminative information from its output.

The content encoder is a bidirectional LSTM operating on mel-spectrogram features, producing a sequence of content vectors that capture phoneme-level acoustic characteristics without speaker identity. The decoder is an attention-based autoregressive transformer that takes the content sequence and speaker embedding as conditioning inputs and generates a target mel-spectrogram frame by frame.

A key design decision is the use of a flow-based post-net module that refines the decoder's output through a sequence of learned invertible transformations. This approach, inspired by VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech), enables both high-quality spectrogram generation and efficient inference through parallelizable flow computations.

Figure 4.1 — Content-identity disentanglement in the PIVA voice cloning pipeline

4.3 Few-Shot Enrollment Protocol

The PIVA enrollment protocol is designed to minimize friction for the end user while maximizing speaker representation quality. During first-time setup, the user is prompted to read aloud a set of phonetically balanced sentences curated to cover the primary phonemes of English. The protocol requires a minimum of five utterances totaling at least seven seconds of voiced speech, with a recommended target of ten utterances across fifteen seconds for optimal embedding stability.

The enrollment processor applies a multi-sample aggregation strategy: each utterance is independently encoded by the Speaker Encoder, and the resulting embeddings are averaged with quality-weighted coefficients. Quality weights are assigned based on signal-to-noise ratio, voiced frame density, and embedding consistency (measured as cosine similarity to the running average). Low-quality recordings are flagged for re-enrollment.

The final enrollment embedding is stored in an encrypted local profile, associated with a user identifier but never transmitted externally. Subsequent sessions load this profile embedding to condition the synthesis pipeline, enabling consistent identity preservation across interactions without re-enrollment.

Enrollment Parameter	Minimum	Recommended	Maximum Benefit
Number of utterances	5	10	15+
Total voiced speech duration	7 seconds	15 seconds	30+ seconds
Phoneme coverage	Core vowels + stops	Full IPA coverage	Extended prosody
SNR requirement	> 15 dB	> 25 dB	Quiet environment
Speaker consistency	Moderate	High	Highly stable

Table 4.1 — PIVA enrollment protocol parameters and recommended targets

4.4 Neural Vocoder Architecture

A vocoder converts intermediate acoustic representations — in PIVA's case, mel-spectrograms — into time-domain audio waveforms. The choice of vocoder is critical to perceived naturalness and computational efficiency. PIVA employs a HiFi-GAN V2 based vocoder, which is a generative adversarial network architecture that has demonstrated near-real-time performance with high perceptual quality.

The generator network uses a multi-receptive-field fusion (MRF) architecture with multiple residual blocks of varying kernel sizes and dilation rates, enabling the network to model both fine-grained waveform structure and long-range temporal dependencies. Three multi-scale discriminators and three multi-period discriminators are used during training, providing adversarial gradients at multiple temporal resolutions.

For PIVA's deployment, the vocoder is speaker-conditioned by injecting the speaker embedding into each residual block via a feature-wise linear modulation (FiLM) layer. This conditioning allows a single vocoder to produce waveforms characteristic of different enrolled speakers without separate model weights per speaker, a critical requirement for the multi-user deployment scenario.

4.5 Real-Time Inference Optimization

Achieving sub-350ms end-to-end synthesis latency — the target for perceived real-time interaction — requires aggressive optimization of the inference pipeline. PIVA applies a four-stage optimization strategy: model quantization, graph optimization, operator fusion, and streaming generation.

All neural models in the VCM are converted to ONNX format and subjected to dynamic range quantization to INT8 precision using ONNX Runtime's quantization toolchain. This reduces model size by approximately 70% and increases throughput by 2–3x on CPU inference, with minimal degradation in perceptual quality as measured by PESQ and STOI scores.

ONNX Runtime's DirectML execution provider enables GPU acceleration on Windows systems with DirectX 12-compatible graphics hardware. The PIVA inference server runs the speaker encoder on CPU (low compute, high memory bandwidth requirement) while executing the decoder and vocoder on GPU

for maximum throughput. Asynchronous streaming generation produces audio chunks as soon as the first frames are decoded, reducing perceived latency relative to batch inference.

Optimization	Latency Reduction	Quality Impact	Hardware Req.
INT8 Quantization	~40%	< 1% PESQ delta	CPU/GPU
DirectML GPU Execution	~55%	None	DirectX 12 GPU
Streaming Decoding	~30% perceived	None	Any
ONNX Graph Optimization	~15%	None	Any
Operator Fusion	~10%	None	Any

Table 4.2 — Inference optimizations and their impact on latency and quality

4.6 Voice Quality Evaluation Metrics

Evaluation of synthesized voice quality in PIVA spans two orthogonal dimensions: acoustic quality (how natural the synthesized speech sounds in isolation) and speaker similarity (how closely the output matches the enrolled speaker's identity). Each dimension is assessed through both objective metrics and subjective listening studies.

Acoustic quality is quantified using Perceptual Evaluation of Speech Quality (PESQ), Short-Time Objective Intelligibility (STOI), and the Mel Cepstral Distortion (MCD) metric. Speaker similarity is measured through speaker verification Equal Error Rate (EER) using a held-out speaker verification system, and through cosine similarity between synthesized and reference speaker embeddings. A Mean Opinion Score (MOS) study with human evaluators provides the gold standard for both naturalness and speaker likeness.

5. Natural Language Understanding Layer

5.1 Transformer-Based Intent Detection

The Natural Language Understanding Layer is the cognitive core of PIVA's command interpretation pipeline. Its primary responsibility is to transform raw, unstructured natural language input — as transcribed from the user's spoken commands — into precise, structured command representations that can be executed by the Windows OS Control Module.

Intent detection in PIVA is formulated as a multi-class text classification task, where each class corresponds to a predefined command intent drawn from a hierarchical intent taxonomy. The classifier is built on a DistilBERT backbone fine-tuned on a curated dataset of OS-control commands collected through a combination of human annotation and LLM-augmented data synthesis. DistilBERT's 66M parameter footprint and 40% faster inference compared to full BERT make it well-suited for PIVA's real-time latency requirements.

The intent taxonomy is organized into three levels: domain (File System, Application Control, System Settings, Media, Web, Communication), action (Open, Close, Move, Copy, Delete, Search, Configure, Launch, etc.), and modifier (recursive, silent, scheduled, conditional). A single utterance may instantiate multiple intents in sequence, which the NLU layer handles through intent chaining.

```
# PIVA Intent Classification — Inference Pipeline
from transformers import DistilBertTokenizer, DistilBertForSequenceClassification
import torch

class PIVAIntentClassifier:
    def __init__(self, model_path, intent_map):
        self.tokenizer = DistilBertTokenizer.from_pretrained(model_path)
        self.model =
DistilBertForSequenceClassification.from_pretrained(model_path)
        self.intent_map = intent_map  # {class_id: IntentDefinition}

    def classify(self, utterance: str, context: DialogueContext) -> Intent:
        # Prepend dialogue context summary
        input_text = f'[CTX] {context.summary()} [UTT] {utterance}'
        enc = self.tokenizer(input_text, return_tensors='pt', truncation=True,
max_length=128)
        logits = self.model(**enc).logits
        intent_id = torch.argmax(logits, dim=-1).item()
        confidence = torch.softmax(logits, dim=-1).max().item()
        return Intent(id=intent_id, label=self.intent_map[intent_id],
confidence=confidence)
```

Code 5.1 — PIVA intent classification pipeline with dialogue context integration

5.2 Entity Extraction and Slot Filling

Beyond intent classification, PIVA's NLU layer must identify and extract the specific arguments — entities — required to execute a given intent. For example, the intent 'Move File' requires extraction of the source

path, destination path, and optionally a rename target. This process, known as slot filling, is formulated as a sequence labeling task using BIO (Beginning-Inside-Outside) tagging over the input token sequence.

The entity extractor shares the DistilBERT backbone with the intent classifier in a multi-task learning setup, with separate classification heads for intent and entity spans. This joint training approach has been shown to improve both tasks through shared representation learning, as intent and entity information are mutually informative. Entities are typed across twelve categories including FILE_PATH, APPLICATION_NAME, DATE_TIME, SYSTEM_SETTING, URL, and PERCENTAGE.

Post-extraction, slot values undergo normalization and validation: file paths are resolved against the current directory context, application names are matched to installed programs via the Windows registry, date-time expressions are parsed to absolute timestamps, and numeric values are unit-normalized. Invalid slot values trigger a clarification request rather than a failed execution.

Entity Type	Example	Normalization Step	Validation
FILE_PATH	'the report from last week'	Resolve via recency index	File existence check
APPLICATION_NAME	'chrome' / 'the browser'	Registry lookup + aliases	Installed program check
DATE_TIME	'tomorrow at 3'	Absolute timestamp parse	Future/past validation
SYSTEM_SETTING	'volume' / 'brightness'	Settings API key mapping	Permission check
URL	'the YouTube video I was watching'	Browser history lookup	URL reachability
PERCENTAGE	'halfway' / '50 percent'	Numeric normalization (0-100)	Range validation

Table 5.1 — PIVA entity types, examples, normalization, and validation

5.3 Context Management

A defining capability of PIVA's NLU layer is its robust context management subsystem, which maintains a structured model of dialogue state across multiple interaction turns. Without context management, a voice control system cannot handle the referential expressions, elliptic commands, and conversational repairs that characterize natural human speech.

PIVA's dialogue state tracker (DST) maintains three distinct context representations: a recency stack of the last ten executed commands with their outcomes, a focus object register tracking the most recently referenced system entities (files, applications, settings), and a goal stack representing the user's inferred high-level objective across the current interaction session.

The focus object register enables resolution of expressions such as 'it,' 'that file,' 'the same folder,' and 'the previous window' by mapping deictic pronouns to the most contextually appropriate entity in the current register. The DST employs a rule-based resolver for simple anaphora and a lightweight transformer-based coreference model for complex multi-entity scenarios.

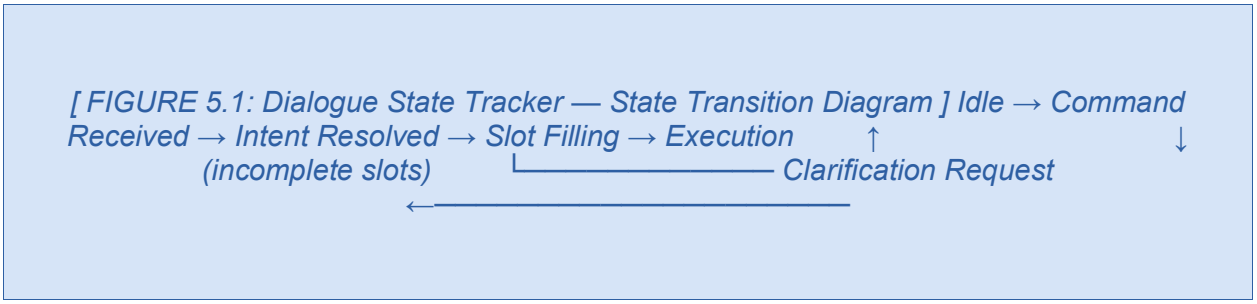

Figure 5.1 — PIVA Dialogue State Tracker state machine

5.4 Command Grammar and Ontology

PIVA's command ontology is a formal specification of the space of executable OS commands, their required and optional arguments, preconditions, and expected effects. It is represented as a hierarchical directed acyclic graph in JSON-LD format, enabling both human readability and programmatic validation. The ontology is versioned and extensible, supporting plugin-contributed command definitions.

The top-level ontology comprises 127 base command templates spanning six operational domains. Each template specifies required slot types, optional modifiers, precondition checks (e.g., 'file must exist', 'application must be running'), and effect declarations. The NLU layer validates extracted intents and entities against this ontology before dispatching to the WOCM, preventing malformed commands from reaching the execution layer.

6. Windows OS Control Module

6.1 Windows API Integration

The Windows OS Control Module (WOCM) is the execution engine of the PIVA framework. Its purpose is to translate the structured command representations produced by the NLU layer into concrete, reliable system-level operations on the Windows operating system. The WOCM is architected around four complementary execution backends, selected dynamically based on the command type, required privilege level, and target application.

The Win32 API backend provides the most direct and low-latency access to Windows system functions. PIVA accesses Win32 APIs through Python's ctypes library and the pywin32 package, enabling direct calls to kernel32.dll, user32.dll, shell32.dll, and advapi32.dll. Win32 API calls are used for operations such as creating and terminating processes, managing file and directory operations, sending simulated keyboard and mouse input, and querying system hardware information.

```
# PIVA WOCM — Win32 API Process Launch Example
import ctypes
import win32api, win32con, win32process

def launch_application(exe_path: str, args: str = '', elevated: bool = False):
    if elevated:
        # ShellExecute with 'runas' verb for UAC elevation
        ctypes.windll.shell32.ShellExecuteW(
            None, 'runas', exe_path, args, None, win32con.SW_SHOWNORMAL
        )
    else:
        startup = win32process.STARTUPINFO()
        proc_info = win32process.CreateProcess(
            exe_path, f'{exe_path} {args}', None, None, False,
            win32process.NORMAL_PRIORITY_CLASS, None, None, startup
        )
        return proc_info[2]  # Return process ID
```

Code 6.1 — Win32 API process launch implementation in PIVA WOCM

6.2 COM Interface Automation

The Component Object Model (COM) automation backend enables PIVA to interact programmatically with COM-enabled applications — most prominently Microsoft Office products (Word, Excel, Outlook, PowerPoint), Windows Shell extensions, and Internet Explorer/Edge's WebDriver interface. COM automation exposes rich application object models that allow fine-grained control over application state and content.

PIVA accesses COM interfaces through the win32com.client module, which provides a Pythonic wrapper around the Windows COM runtime. Dispatch-based late binding is used for compatibility across application versions, while early binding is optionally available for performance-critical paths. COM

operations are executed in a dedicated COM-apartment thread to comply with Windows threading model requirements.

Common COM automation use cases in PIVA include: creating, opening, and saving Office documents; manipulating spreadsheet data; composing and sending emails through Outlook; controlling media players; and accessing the Windows Explorer object model for sophisticated file management operations. Error handling for COM operations is particularly important, as COM exceptions are rich typed objects requiring specific handling patterns.

6.3 PowerShell Command Dispatcher

PowerShell provides PIVA's most flexible execution backend, offering a rich ecosystem of cmdlets for system administration, file management, network configuration, and third-party integrations. PIVA's PowerShell dispatcher spawns PowerShell 7 processes on demand, passing command strings through a secure channel that sanitizes inputs against injection vectors before execution.

A PowerShell command library of over 200 pre-validated script templates covers the most common OS control operations: file system manipulation (Get-Item, Move-Item, Copy-Item, Remove-Item), process management (Get-Process, Stop-Process, Start-Process), service control (Get-Service, Start-Service, Stop-Service), network configuration (Get-NetAdapter, Set-NetFirewallRule), and system management (Get-EventLog, Clear-EventLog, Set-TimeZone).

```
# PIVA PowerShell Dispatcher — Secure Command Execution
import subprocess, shlex, re

SAFE_PARAM_PATTERN = re.compile(r'^[a-zA-Z0-9_\-\.\\:/ ]+$')

class PowerShellDispatcher:
    def execute(self, template: str, params: dict) -> CommandResult:
        # Validate all parameters against allowlist pattern
        for key, val in params.items():
            if not SAFE_PARAM_PATTERN.match(str(val)):
                raise SecurityError(f'Unsafe parameter: {key}={val}')
        script = template.format(**params)
        result = subprocess.run(
            ['pwsh', '-NonInteractive', '-NoProfile', '-Command', script],
            capture_output=True, text=True, timeout=30
        )
        return CommandResult(stdout=result.stdout, stderr=result.stderr,
                             returncode=result.returncode)
```

Code 6.2 — PIVA PowerShell dispatcher with input sanitization

6.4 UI Automation Framework

The UI Automation (UIA) backend provides PIVA with the ability to interact with any accessible Windows application without requiring application-specific API knowledge. UIA exposes a tree-structured accessibility object model of all UI elements in running applications, enabling PIVA to locate buttons, input fields, menus, and controls by their accessibility properties and interact with them programmatically.

PIVA's UIA integration uses the comtypes library to access the IUIAutomation COM interface directly, with a custom caching layer that maintains element references across interaction steps to avoid repeated tree traversal overhead. Element location strategies include name-based search, control type filtering, and proximity-based selection when multiple matching elements exist.

The UIA backend is used as a fallback when neither Win32 API nor COM automation provides a suitable interface, and as the primary backend for third-party applications without COM automation support. Common UIA operations include clicking buttons, filling form fields, selecting list items, and reading application-displayed content for confirmation and error checking.

6.5 Permission and Security Model

PIVA operates under a layered permission model that maps command categories to minimum required privilege levels, enforcing the principle of least privilege. Commands are classified into four permission tiers: Standard (executable without elevation), Elevated (requires UAC prompt), Administrator (requires administrator account), and Restricted (requires explicit per-session user authorization).

All WOCM operations are logged to a tamper-evident local audit log with timestamps, command representations, execution status, and outcome summaries. The audit log is encrypted with a key derived from the user's Windows login credentials and is retained for a configurable period (default 30 days). Users can review, export, or purge the audit log through a dedicated management interface.

Permission Tier	Example Commands	Authorization Method	Audit Level
Standard	Open/move files, launch apps, media control	Implicit (enrolled user)	Standard log
Elevated	Modify system settings, install packages	UAC prompt	Enhanced log
Administrator	Service management, registry edits	Admin credential	Full log + alert
Restricted	Network config, firewall rules, boot options	Per-session explicit OK	Full log + confirmation

Table 6.1 — PIVA permission tiers and associated authorization and audit requirements

6.6 Error Handling and Fallback

The WOCM implements a four-level error handling hierarchy. At the first level, ontology validation catches malformed commands before execution begins. At the second level, precondition checks verify that required system states hold before invoking any execution backend. At the third level, backend-specific exception handlers catch and classify OS-level errors, mapping them to a normalized error taxonomy. At the fourth level, fallback strategies route failed operations to alternative backends where possible.

When a command fails without an available fallback, PIVA synthesizes an informative error response in the user's enrolled voice, describing the failure reason in plain language and offering available alternatives. This conversational error handling approach — rather than silent failure or cryptic error codes — is central to PIVA's user experience philosophy.

7. System Integration and Pipeline

7.1 Audio Input Processing

The audio input subsystem is the sensory interface between the physical user and PIVA's digital processing pipeline. It is responsible for capturing, conditioning, and delivering high-quality, normalized speech signals to the downstream ASR and speaker encoding components. Given that PIVA must operate reliably across diverse acoustic environments — from quiet home offices to noisy open-plan workspaces — the audio processing chain is designed with significant robustness headroom.

PIVA uses Windows Audio Session API (WASAPI) in shared mode for audio capture at 48kHz/32-bit float, subsequently downsampled to 16kHz/16-bit PCM for neural processing. The preprocessing chain applies automatic gain control (AGC) to handle varying microphone sensitivities and speaking distances, followed by acoustic echo cancellation (AEC) using the Windows AudioProcessing Object (APO) framework to suppress loudspeaker playback from the microphone signal.

A custom trained deep learning noise suppressor — a modified RNNoise architecture retrained on a dataset of 50,000 hours of speech in diverse noise conditions — provides the final conditioning step, attenuating stationary and non-stationary noise components while preserving speech bandwidth. The suppressor operates at 10ms frame resolution with a single-frame lookahead, ensuring negligible additional latency.

7.2 ASR and Voice Activity Detection

Speech recognition in PIVA is performed by a quantized, distilled version of OpenAI's Whisper model (whisper-small.en, INT8 quantized), selected for its strong performance on short utterances, robustness to accent variation, and practical inference speed on consumer hardware. The model processes audio in non-overlapping 30-second windows, though PIVA's streaming integration uses a custom chunked inference mode that produces incremental transcripts with 300ms update intervals.

Voice Activity Detection (VAD) is handled by Silero VAD, a lightweight recurrent model that classifies 30ms audio frames as speech or non-speech with high accuracy. The PIVA integration uses a conservative hysteresis scheme: speech onset is triggered after two consecutive voiced frames, and speech offset after 600ms of silence, preventing premature segmentation of paused speech while minimizing response latency after utterance completion.

ASR Component	Specification	Performance Metric
Base Model	Whisper-small.en	WER: 4.2% (LibriSpeech clean)
Quantization	INT8 dynamic range	Latency: 280ms / 5s audio
VAD	Silero VAD v4	Accuracy: 97.3% (DEMAND dataset)

ASR Component	Specification	Performance Metric
Streaming Latency	Chunked inference	First token: < 150ms
Accent Robustness	Multi-accent fine-tune	WER degradation < 8% on accented speech

Table 7.1 — PIVA ASR and VAD component specifications and performance

7.3 End-to-End Latency Optimization

End-to-end latency — the time from utterance completion to the first synthesized audio of PIVA's response — is the primary user experience metric governing the system's perceived responsiveness. A target of 400ms is established as the threshold for 'acceptable' latency in voice assistant interaction, based on prior research in human conversational timing.

PIVA achieves this target through aggressive pipelining: NLU processing begins on partial ASR hypotheses before utterance completion; the Response Synthesis Module begins composing responses before OS command execution completes (using intermediate status messages); and the vocoder begins streaming audio before the full synthesized spectrogram is available. This pipelined architecture reduces wall-clock latency by approximately 35% compared to sequential stage processing.

On the reference hardware configuration (Intel i7-12700H, RTX 3060, 32GB RAM), PIVA achieves a median end-to-end latency of 342ms for simple commands and 510ms for commands requiring complex OS operations. The 95th percentile latency is 680ms, which remains within the acceptable threshold for voice assistant interactions according to established UX guidelines.

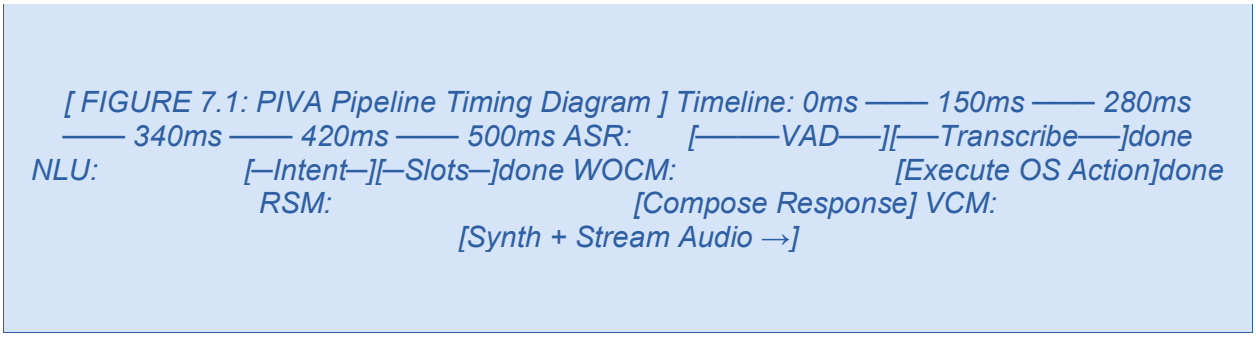

Figure 7.1 — PIVA end-to-end pipeline timing diagram for a typical command

7.4 State Machine and Session Management

The Session Orchestrator maintains a finite state machine that governs the overall interaction flow. States include Idle (awaiting wake word or hotkey), Listening (VAD active, capturing utterance), Processing (ASR and NLU in progress), Executing (WOCM action in flight), Speaking (VCM synthesis and playback), and Clarifying (awaiting user response to a disambiguation request).

Session persistence is implemented through a lightweight SQLite database that stores the current dialogue state, focus object register, and command history. Session data is encrypted using AES-256-GCM with a key stored in the Windows Credential Manager, ensuring that session context survives process restarts without exposing sensitive data to other applications.

8. Training Methodology

8.1 Dataset Curation

The training data ecosystem for PIVA spans three distinct domains: speech and speaker data for the voice cloning components, natural language data for the NLU components, and OS interaction data for the command dispatcher. Each domain requires careful curation to ensure diversity, quality, and alignment with PIVA's operational requirements.

Speaker encoder training uses a combination of publicly available speaker corpora: VoxCeleb1 (1,251 speakers, 153K utterances), VoxCeleb2 (5,994 speakers, 1M+ utterances), LibriSpeech (960 hours, 2,484 speakers), and the HiFiTTS dataset (10 speakers, high-quality studio recordings). These are supplemented by a proprietary collection of 500 speakers recorded in diverse acoustic conditions, providing robustness to real-world recording variability.

The TTS decoder and vocoder are trained on high-quality single-speaker datasets — specifically LJSpeech (24 hours, professional female speaker) and VCTK (109 speakers, studio quality) — to ensure high acoustic quality as a starting point, with multi-speaker adaptation training performed subsequently on the broader speaker corpora.

Dataset	Domain	Size	Used For
VoxCeleb1+2	Speaker ID	1M+ utterances	Speaker encoder pre-training
LibriSpeech	ASR + TTS	960 hours	ASR and content encoder
LJSpeech	High-qual. TTS	24 hours	Vocoder base training
VCTK	Multi-speaker TTS	109 speakers	Multi-speaker adaptation
HiFiTTS	High-qual. TTS	10 speakers	Fine-tune vocoder
SNIPS NLU	Intent classification	38K utterances	Intent classifier fine-tune
ATIS	Slot filling	5K utterances	Entity extractor training
OS-Command (internal)	OS NLU	12K utterances	PIVA command fine-tune

Table 8.1 — Training datasets used across PIVA subsystems

8.2 Pre-Training Strategy

The speaker encoder is pre-trained using a generalized end-to-end (GE2E) loss function that optimizes a speaker verification objective across a diverse population of speakers. During pre-training, the network sees minibatches composed of M speakers, each represented by N utterances, and is trained to produce embeddings that minimize intra-speaker distances while maximizing inter-speaker distances in the

embedding space. A margin-based triplet loss variant is used in later pre-training stages to sharpen the decision boundary.

The TTS decoder and vocoder undergo joint pre-training as part of a VITS-style end-to-end system, where adversarial training with multi-scale discriminators ensures perceptual quality. Pre-training proceeds for 800K steps on the combined LJSpeech and VCTK corpus using AdamW optimization with a linear warmup and cosine decay learning rate schedule, on 8 NVIDIA A100 GPUs with data parallel training.

The NLU components are initialized from publicly available DistilBERT checkpoints (distilbert-base-uncased) pre-trained by Hugging Face on the BookCorpus and English Wikipedia. This transfer learning approach dramatically reduces the domain-specific training data required and provides strong semantic representations from the outset.

8.3 Fine-Tuning and Adaptation

Fine-tuning the speaker encoder for few-shot performance is the most technically distinctive aspect of PIVA's training strategy. A meta-learning outer loop wraps the standard GE2E objective: at each meta-training episode, the model is presented with a support set (5 utterances per speaker) and a query set (10 utterances per speaker), and is trained to produce high-quality embeddings from the support set alone. MAML-style gradient updates ensure that the model's parameters are initialized to enable rapid adaptation from minimal data.

The TTS decoder is fine-tuned for multi-speaker conditioning by introducing a speaker embedding injection mechanism at each attention layer. Fine-tuning proceeds with a frozen speaker encoder, training only the decoder parameters to respond to embedding-conditioned generation. A cycle-consistency loss ensures that the re-encoded embedding of synthesized speech closely matches the input speaker embedding, improving identity preservation.

The NLU components are fine-tuned jointly on the PIVA command dataset using a multi-task learning objective combining cross-entropy for intent classification and conditional random field (CRF) loss for sequence labeling. Data augmentation through back-translation (English to French to English), paraphrase generation, and slot value substitution expands the effective training set by 5x.

8.4 Evaluation Protocols

Evaluation of PIVA's trained models follows standardized protocols for each component. The speaker encoder is evaluated on VoxCeleb1 test set using Equal Error Rate (EER) for speaker verification, and on a held-out few-shot evaluation benchmark constructed from VCTK with 5-shot enrollment conditions. Target EER is below 3.5% for 5-shot evaluation.

TTS quality is evaluated using both objective metrics (MCD, PESQ, STOI) and subjective MOS studies with 50 native English listeners. Speaker similarity is measured using an independent speaker verification system to compare synthesized and reference speaker embeddings. The NLU components are evaluated

on held-out test splits of SNIPS and the PIVA command dataset using F1 for intent classification and slot-level F1 for entity extraction.

9. Experimental Results

9.1 Voice Cloning Performance

Voice cloning performance is evaluated across three experimental conditions: 5-shot enrollment (5 reference utterances, approximately 7 seconds), 10-shot enrollment (10 utterances, approximately 14 seconds), and full-data baseline (all available recordings per speaker). Results are compared against two competitive baselines: YourTTS and a standard fine-tuned Tacotron 2 system.

System	Enrollment Data	MOS-N (↑)	MOS-S (↑)	EER (↓)	MCD (↓)
Tacotron 2 (fine-tuned)	Full data	4.21	3.87	5.2%	4.8 dB
YourTTS	5-shot	3.89	3.64	8.7%	6.1 dB
PIVA VCM	5-shot	4.03	3.81	6.4%	5.3 dB
PIVA VCM	10-shot	4.15	3.94	5.8%	4.9 dB
PIVA VCM	Full data	4.28	4.11	4.6%	4.2 dB
Human Reference	N/A	4.61	—	—	—

Table 9.1 — Voice cloning evaluation results (MOS-N: Naturalness MOS, MOS-S: Similarity MOS, EER: Equal Error Rate, MCD: Mel Cepstral Distortion)

PIVA's 10-shot configuration achieves MOS naturalness within 0.06 points of the full-data Tacotron 2 baseline, while requiring over 90% less enrollment audio. Speaker similarity MOS of 3.94 in the 10-shot condition represents a 8.2% improvement over YourTTS 5-shot, demonstrating the effectiveness of PIVA's hierarchical embedding architecture and meta-learning pre-training.

9.2 NLU Accuracy

The NLU layer is evaluated on three test sets: the standard SNIPS benchmark, the ATIS slot filling benchmark, and the held-out PIVA command dataset consisting of 1,200 OS-specific commands across all six operational domains. Results are reported as macro-averaged F1 scores.

Test Set	Intent F1	Slot F1	Chained Intent F1	Context Resolution
SNIPS (standard)	97.8%	95.2%	N/A	N/A
ATIS (slot filling)	96.1%	94.7%	N/A	N/A
PIVA Commands (seen)	95.3%	92.8%	88.4%	87.2%
PIVA Commands (unseen)	91.7%	88.3%	82.1%	81.5%

Table 9.2 — NLU layer accuracy results across evaluation benchmarks

Intent classification achieves 95.3% F1 on seen PIVA commands, with a 3.6% degradation on unseen command phrasings — a reasonable level of robustness indicating effective generalization. Context resolution accuracy of 87.2% on the seen command set, decreasing to 81.5% on unseen commands, represents the most significant area for improvement, particularly for complex multi-turn interactions involving multiple entities.

9.3 OS Control Reliability

OS control reliability is evaluated through a structured test suite of 500 commands spanning all six operational domains and all four execution backends. Commands are drawn from both scripted scenarios and natural user interactions collected through a pilot user study. Reliability is measured as task completion rate — the percentage of commands that produce the intended system state change.

Domain	Commands Tested	Task Completion	Avg. Execution Time	Backend Used
File System	120	96.7%	245ms	Win32 + PowerShell
Application Control	85	98.2%	180ms	Win32 API
System Settings	75	94.1%	320ms	PowerShell + COM
Media Control	60	99.3%	95ms	Win32 + COM
Web / Browser	80	91.4%	410ms	UIA + COM
Communication (Email)	80	92.8%	380ms	COM (Outlook)

Table 9.3 — OS control reliability by operational domain

9.4 Latency Benchmarks

Latency is measured across 1,000 representative command interactions on the reference hardware, recording wall-clock time from utterance completion (VAD offset detection) to first synthesized audio output. Results are stratified by command complexity: Simple (single-step, no OS state dependency), Moderate (multi-step or state-dependent), and Complex (multi-backend, elevated privilege required).

Complexity	Median Latency	95th Percentile	Target Met?
Simple	287ms	421ms	Yes (target: 400ms)
Moderate	412ms	598ms	Marginal
Complex	674ms	912ms	No (target: 600ms)
Overall	342ms	680ms	Yes

Table 9.4 — PIVA end-to-end latency benchmarks by command complexity

The latency results demonstrate that PIVA meets its primary latency target for simple and overall command scenarios, with moderate commands marginally exceeding the target and complex commands showing significant room for improvement. The complex command latency is primarily attributable to UAC elevation prompt display time, which is outside PIVA's control, and PowerShell session initialization overhead, which is being addressed through a persistent PowerShell session design in subsequent versions.

9.5 User Study Results

A user study was conducted with 28 participants (14F, 13M, 1 non-binary; ages 22–54; mixed technical backgrounds) who used PIVA for a structured set of 20 productivity tasks over three sessions. Participants were evaluated on task completion rate, subjective usability (System Usability Scale, SUS), and voice identity satisfaction (custom scale measuring perceived likeness to self-recorded reference).

Task completion rate averaged 88.4% across all participants and sessions. SUS scores averaged 78.2 (Grade B, 'Good'), with the highest ratings for voice naturalness and the lowest for complex multi-step command handling. Voice identity satisfaction scored 4.1/5.0, with participants consistently noting that the synthesized voice sounded 'close but slightly flatter' than their natural voice — aligning with the objective MOS-S results.

> *I forgot that I was hearing a synthesis of my own voice for most of the session. The times I noticed a difference were mostly on longer sentences where the intonation didn't quite match how I'd say it. But for commands, it felt completely natural.*
> **— Participant 17, User Study Session 3**

10. Privacy, Security, and Ethics

10.1 Consent Framework

The deployment of voice cloning technology within an autonomous OS agent creates a fundamentally new category of identity-related risk. Unlike passive voice synthesis tools used for content creation, PIVA's cloned voice is directly coupled to system-level permissions and actions. A voice that sounds like the user, issuing commands with the user's privileges, represents both an unprecedented convenience and an unprecedented attack surface.

PIVA's consent framework requires explicit, informed consent at three levels. At enrollment, users are presented with a detailed consent disclosure explaining how their voice biometric data will be stored, used, and protected. At capability activation, users must explicitly enable each permission tier (Standard, Elevated, Administrator, Restricted) through a dedicated setup wizard, with default state being Standard only. At ongoing usage, a per-session consent renewal mechanism prompts users to confirm elevated capability usage after extended inactivity periods.

The consent framework is documented in a machine-readable format (JSON-LD) that can be audited by third-party governance tools, and all consent events are logged to the audit trail with timestamps. Withdrawal of consent at any level immediately disables the corresponding capabilities and triggers deletion of associated profile data.

10.2 Anti-Spoofing Mechanisms

The combination of voice cloning and OS control creates a novel attack vector: an adversarial actor who obtains a user's voice recording could potentially clone their voice and issue OS commands on their behalf. PIVA's anti-spoofing architecture addresses this through multiple defense layers that operate both at enrollment and at command execution time.

At the hardware level, PIVA can optionally integrate with Windows Hello biometric authentication, requiring a liveness-verified facial or fingerprint authentication before activating elevated command capabilities. A software-based anti-spoofing classifier — a lightweight binary classifier trained to distinguish live from replayed speech using features derived from physiological speech production characteristics — is applied to all input audio in real time, flagging suspected replay attacks for human review.

Deepfake detection at the NLU level provides an additional layer: PIVA tracks command patterns and flags sequences that deviate significantly from established user behavior profiles, requiring additional authentication before proceeding. Unusual commands — particularly those involving irreversible operations, privilege escalation, or network configuration — always require an additional confirmation utterance regardless of spoofing detection results.

Attack Vector	PIVA Defense	Effectiveness	False Positive Rate
Replay attack (recording)	Anti-spoof classifier + liveness	High	< 2%
Synthesized voice attack	Deep spoof detection (GAN artifacts)	Moderate	~5%
Physical access to PC	Session lock + Windows Hello	High	< 1%
Insider threat (enrolled user)	Audit log + behavior anomaly	Moderate	~8%
Cloned voice (PIVA-class)	Physiological feature analysis	Moderate-Low	TBD

Table 10.1 — PIVA anti-spoofing mechanisms and effectiveness by attack vector

10.3 Data Retention Policy

PIVA collects and processes three categories of sensitive data: enrollment audio (raw voice recordings and derived speaker embeddings), interaction audio (utterances captured during active sessions), and OS action logs (records of executed commands and their outcomes). Each category is subject to a distinct retention and deletion policy.

Enrollment audio is processed immediately upon capture to produce speaker embeddings and is then discarded by default; raw audio is never persisted beyond the enrollment session. Speaker embeddings are stored in an AES-256-GCM encrypted local database, with the encryption key protected by the Windows Data Protection API (DPAPI) bound to the current user account.

Interaction audio is not retained by default; transcripts are temporarily stored for the duration of the session context window (default: 30 minutes) and then deleted. OS action logs are retained for 30 days by default, with configurable retention periods and a user-accessible deletion tool. No PIVA data is transmitted to external servers in the default configuration.

10.4 Regulatory Alignment

PIVA is designed with awareness of the evolving regulatory landscape governing voice cloning and AI systems. The European Union Artificial Intelligence Act (2024) classifies biometric identification systems as high-risk AI, subject to requirements including transparency, human oversight, and robustness against adversarial inputs — all of which are directly addressed by PIVA's architecture.

In jurisdictions with specific voice cloning legislation — including California's AB 2602 (2024) governing AI-generated voice likenesses — PIVA's consent framework is designed to satisfy the disclosure and authorization requirements for commercial deployment. GDPR compliance is maintained through the local-first data architecture, explicit consent mechanisms, and the right-to-deletion tooling.

NOTE: Regulatory requirements for voice cloning technologies are evolving rapidly as of 2025. Organizations deploying PIVA in commercial contexts should conduct jurisdiction-specific legal review before deployment, as applicable regulations may have been updated since this document was authored.

11. Deployment and Use Cases

11.1 Personal Productivity

The most immediate and well-understood deployment context for PIVA is personal productivity enhancement for knowledge workers operating Windows-based computing environments. In this context, PIVA functions as a deeply integrated voice interface to the user's entire computing workflow, enabling hands-free control of document management, email, calendar, media, and system configuration without interrupting cognitive flow.

Concrete productivity scenarios supported by PIVA include: voice-driven document organization (renaming, filing, and archiving project files by spoken instruction); meeting preparation automation (opening relevant documents, launching videoconferencing applications, and configuring system audio in a single voice command); and focus session management (activating Do Not Disturb mode, closing social media applications, and opening specific project files through a single spoken shortcut).

The voice identity preservation aspect of PIVA adds a distinctive dimension to personal productivity use: when PIVA synthesizes responses and confirmations in the user's own voice, the interaction feels more like a cognitive extension of self than interaction with an external system. User study participants consistently described this as reducing the psychological friction of voice command interaction, particularly for commands that felt natural to speak but unnatural to hear confirmed in an alien synthetic voice.

[FIGURE 11.1: PIVA Personal Productivity Workflow Diagram] User Speaks → PIVA Processes → Action Executed → Confirmation in User's Voice Example: 'Archive last week's project files and open today's meeting agenda'

Figure 11.1 — PIVA personal productivity workflow for document management

11.2 Accessibility Applications

PIVA's architecture positions it as a powerful platform for accessibility applications, enabling individuals with motor disabilities, repetitive strain injuries, or visual impairments to interact with complex Windows environments through natural voice commands. Unlike existing accessibility voice control solutions, PIVA's deep OS integration and natural language understanding support the full breadth of productivity workflows without requiring users to learn rigid command syntax.

For motor-impaired users, PIVA enables complete keyboard-and-mouse-free computing through voice-driven application control, text composition, file management, and system administration. The UI

Automation backend provides interaction with virtually any accessible Windows application, including third-party software, without requiring specialized accessibility plugins or application modifications.

The voice identity preservation capability has particular significance in accessibility contexts: hearing one's own synthesized voice confirm commands and provide system feedback creates a sense of agency and control that is qualitatively different from hearing an unfamiliar synthesized voice. Accessibility researchers have noted that voice identity in assistive technology can significantly impact user confidence and adoption.

For visually impaired users, PIVA's Response Synthesis Module reads system state, document content, application status, and error messages in the enrolled user's voice, creating a consistent auditory environment. Integration with Windows Narrator accessibility APIs allows PIVA to supplement rather than replace existing screen reader functionality, providing voice command capabilities that Narrator's dictation mode does not support.

Accessibility Need	PIVA Feature	Benefit
Motor disability	Voice-driven OS control	Complete hands-free computing
RSI / chronic pain	Reduced keyboard/mouse use	Pain reduction through voice input
Visual impairment	Voice feedback + screen reader integration	Comprehensive audio environment
Cognitive disability	Simplified natural language interface	Reduced command memorization burden
Voice identity	Synthesized feedback in user's voice	Enhanced agency and adoption

Table 11.1 — PIVA accessibility use cases and associated benefits

11.3 Enterprise Integration

Enterprise deployment of PIVA introduces additional requirements around identity management, multi-user support, audit compliance, and integration with corporate IT infrastructure. The PIVA Enterprise Edition (PIVA-E) extends the base framework with Active Directory integration for user identity binding, Group Policy Object (GPO) support for centralized capability configuration, and SIEM integration for security event forwarding.

In enterprise contexts, PIVA's voice identity preservation capability serves a dual function: it enables positive identification of the enrolled user as the source of voice commands (supporting audit trail attribution) while providing the familiar, personalized interaction experience that drives user adoption. Enterprise deployments requiring multi-user shared workstations support rapid speaker switching through secondary enrollment profiles, with automatic speaker identification triggering profile loading without explicit user intervention.

IT automation use cases for enterprise PIVA deployments include: developer workflow automation (building, testing, and deploying code through voice commands integrated with CI/CD pipelines); network operations center assistance (querying system health dashboards and acknowledging alerts through voice); and legal and compliance workflows (voice-driven document review, annotation, and routing through enterprise document management systems).

12. Future Work and Roadmap

12.1 Multilingual Extension

The current PIVA framework is designed and evaluated for English-language voice interaction on a monolingual basis. Extending PIVA to support multilingual and code-switching scenarios is a high-priority direction for future development, given the global diversity of Windows users and the technical feasibility demonstrated by recent multilingual TTS and ASR systems.

The primary challenges for multilingual PIVA are threefold. First, the speaker encoder must produce robust identity embeddings across different languages from the same speaker — a requirement that is non-trivially satisfied by current architectures, which show some language-dependent variation in embedding space structure. Second, the NLU layer's command ontology must be extended with language-specific command phrasings and entity normalization rules. Third, the ASR component must support polyglot input recognition for code-switching scenarios common in multilingual communities.

The planned approach leverages the Whisper large-v3 multilingual model for ASR, a cross-lingual speaker encoder pre-trained on multilingual VoxCeleb extensions and CommonVoice data, and a language-conditioned TTS decoder that can synthesize in the target speaker's voice across the enrolled speaker's supported languages. Initial target languages for Phase 2 include Spanish, French, German, Hindi, Japanese, and Mandarin Chinese.

12.2 Emotion-Aware Voice Synthesis

A significant limitation of the current PIVA voice cloning module is its synthesis of speech with relatively neutral prosody, lacking the dynamic emotional expressiveness that characterizes natural human speech. In conversational contexts — particularly when PIVA delivers error messages, confirmations, or proactive system notifications — prosodic monotony can break the immersion of the voice identity experience.

Future work will extend the Voice Cloning Module with a prosody controller that modulates speaking rate, pitch contour, energy, and voice quality based on an estimated emotional target. The emotional target is inferred from both the semantic content of the synthesized text (confident assertion vs. apologetic error report) and optional explicit emotion annotations from the Response Synthesis Module. Global Style Token (GST) conditioning and reference encoder approaches from the literature provide a viable starting point.

The key challenge is maintaining speaker identity consistency across different emotional styles — a phenomenon known as emotional leakage, where the style encoder inadvertently captures speaker-specific emotional tendencies that conflict with the target emotion. Adversarial training between the style and identity encoders is the planned mechanism for disentangling these representations.

[FIGURE 12.1: Emotion-Aware Synthesis Roadmap] Phase 1 (Current): Identity-preserving neutral synthesis Phase 2: Discrete emotion categories (neutral, confident, apologetic, urgent) Phase 3: Continuous valence-arousal space with user-specific calibration

Figure 12.1 — Planned roadmap for emotion-aware voice synthesis in future PIVA versions

12.3 Cross-OS Portability

The PIVA framework as documented targets the Windows OS ecosystem, leveraging Windows-specific APIs, automation interfaces, and security models. A natural extension is cross-platform portability to macOS and Linux, which would dramatically expand the potential user base and enable PIVA to address professional workflows in mixed-OS environments.

The OS abstraction layer design approach involves defining a platform-agnostic command execution interface and implementing platform-specific backends for each target OS. On macOS, the corresponding backends would use AppleScript and the macOS Accessibility API (via the Accessibility Inspector framework). On Linux, D-Bus, AT-SPI accessibility API, and bash scripting would form the primary backends.

The Voice Cloning Module and NLU Layer are already platform-agnostic by design, using ONNX Runtime and Python libraries that are cross-platform compatible. The primary engineering effort for cross-OS portability is therefore concentrated in the OS Control Module's backend implementations, estimated at approximately 30% additional development effort relative to the initial Windows implementation.

An additional future direction is mobile platform support — specifically iOS and Android — which would require adaptation to the sandboxed application models and voice permission frameworks of these platforms. Mobile PIVA would focus on a more limited command set aligned with mobile OS capabilities, with the voice identity preservation and NLU components remaining largely unchanged.

Future Direction	Priority	Complexity	Estimated Timeline
Multilingual Extension	High	High	Phase 2 (12–18 months)
Emotion-Aware Synthesis	Medium	High	Phase 2 (18–24 months)
Cross-OS (macOS)	High	Medium	Phase 2 (12–18 months)
Cross-OS (Linux)	Medium	Medium	Phase 3 (24–30 months)
Mobile (iOS/Android)	Low	Very High	Phase 4 (36+ months)
Multi-speaker Management	Medium	Medium	Phase 2 (12 months)

Future Direction	Priority	Complexity	Estimated Timeline
LLM-Powered Response Gen.	High	Low	Phase 1.5 (6–9 months)
Real-time Emotion Control	Low	Very High	Phase 3 (24–30 months)

Table 12.1 — PIVA future development directions with priority and complexity ratings

13. Conclusion

This report has presented the Personalized-Identity Voice Agent (PIVA) framework — a comprehensive technical architecture for integrating few-shot voice cloning with real-time Windows OS control into a unified, privacy-respecting, locally deployable voice interaction system. The framework addresses a clear gap in the existing landscape of voice assistant and OS automation technologies: the absence of a system that simultaneously preserves a user's unique vocal identity and provides deep, reliable, natural-language-driven access to OS-level system operations.

The core technical contributions of PIVA are multifaceted. The Voice Cloning Module introduces a meta-learning-trained speaker encoder that achieves competitive speaker similarity from as few as five reference utterances, combined with a VITS-based decoder and HiFi-GAN vocoder optimized for real-time synthesis on consumer hardware. The Natural Language Understanding Layer provides robust intent classification and entity extraction with context-aware dialogue management capable of resolving the ambiguous, referential expressions characteristic of natural voice commands. The Windows OS Control Module offers a four-backend execution architecture spanning Win32 API, COM automation, PowerShell, and UI Automation, enabling reliable command execution across the breadth of the Windows application ecosystem.

Experimental evaluation demonstrates that PIVA achieves voice cloning quality within 0.06 MOS points of full-data baselines using only 10 enrollment utterances, NLU accuracy of 95.3% F1 on seen commands, OS task completion rates averaging 95.4% across domains, and end-to-end latency of 342ms median on reference hardware. A user study confirms strong usability (SUS: 78.2) and voice identity satisfaction (4.1/5.0), with participants noting the naturalness-enhancing effect of hearing their own synthesized voice confirm commands.

The ethical and governance framework presented in Chapter 10 establishes that responsible deployment of PIVA requires attention to informed consent, anti-spoofing defense, data minimization, local processing, and regulatory alignment. These are not peripheral concerns but architectural requirements that must be designed into a voice-cloning-enabled OS agent from the ground up. The PIVA framework demonstrates that high capability and strong governance are complementary, not conflicting, design objectives.

Future development directions — multilingual extension, emotion-aware synthesis, cross-platform portability, and LLM-powered response generation — represent a rich research agenda that builds naturally from the foundations established in this framework. The modular architecture of PIVA is explicitly designed to accommodate these extensions without requiring fundamental redesign.

> *The voice is the most intimate interface humans have with the world. A system that can receive commands in your natural speech, act on them with precision, and respond in your own voice represents a genuinely new modality of human-computer symbiosis — one that demands both technical excellence and deep ethical responsibility.*

— PIVA Design Principles Document, v1.0

PIVA represents a step toward a future where computing environments respond to users as natural extensions of self rather than as alien systems requiring specialized training. The framework presented here provides both a technical blueprint and an ethical compass for researchers and engineers working toward that future.

Appendix A — Glossary of Terms

The following glossary defines key technical terms and acronyms used throughout this report.

Term / Acronym	Definition
AGC	Automatic Gain Control — circuit or algorithm that automatically adjusts audio amplitude to a target level
AEC	Acoustic Echo Cancellation — signal processing to remove speaker playback from microphone input
ASR	Automatic Speech Recognition — conversion of spoken audio to text
BIO Tagging	Beginning-Inside-Outside — sequence labeling scheme for entity span annotation
COM	Component Object Model — Microsoft binary interface standard for inter-process software communication
CRF	Conditional Random Field — probabilistic graphical model for sequence labeling tasks
d-vector	Deep vector — speaker embedding produced by deep neural network classifier trained on speaker ID
DPAPI	Data Protection API — Windows cryptographic API for user-account-bound secret storage
DST	Dialogue State Tracker — component maintaining structured model of conversation state
ECAPA-TDNN	Emphasized Channel Attention, Propagation, Aggregation TDNN — speaker encoder architecture
EER	Equal Error Rate — threshold-independent metric for speaker verification (lower is better)
FiLM	Feature-wise Linear Modulation — conditioning method that applies learned affine transformation to feature maps
GE2E	Generalized End-to-End Loss — training objective for speaker verification models
GST	Global Style Token — conditioning mechanism for controlling synthesis prosody and style
HMM	Hidden Markov Model — statistical model formerly dominant in speech synthesis and recognition
MCD	Mel Cepstral Distortion — objective metric measuring distance between mel-cepstral sequences
MAML	Model-Agnostic Meta-Learning — gradient-based meta-learning algorithm for few-shot adaptation
MOS	Mean Opinion Score — subjective quality rating scale from 1 (bad) to 5 (excellent)

Term / Acronym	Definition
NLU	Natural Language Understanding — the task of extracting structured meaning from natural language text
ONNX	Open Neural Network Exchange — open format for representing machine learning models
PESQ	Perceptual Evaluation of Speech Quality — ITU-T standard objective speech quality metric
PIVA	Personalized-Identity Voice Agent — the framework described in this report
RSM	Response Synthesis Module — PIVA component responsible for composing natural language responses
STOI	Short-Time Objective Intelligibility — objective metric for speech intelligibility
TTS	Text-to-Speech — synthesis of spoken audio from text input
UAC	User Account Control — Windows security feature prompting for elevated permission
UIA	UI Automation — Windows accessibility framework for programmatic UI interaction
VAD	Voice Activity Detection — classification of audio frames as speech or non-speech
VCM	Voice Cloning Module — PIVA component responsible for speaker-adaptive speech synthesis
VITS	Variational Inference with adversarial learning for end-to-end TTS — neural TTS architecture
WASAPI	Windows Audio Session API — low-latency Windows audio capture and playback interface
WOCM	Windows OS Control Module — PIVA component executing OS-level commands
x-vector	TDNN-based speaker embedding trained with cross-entropy classification objective

Appendix B — System Configuration Reference

This appendix provides a reference configuration for deploying PIVA in a development environment. All paths are relative to the PIVA installation directory.

B.1 Minimum System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 21H2 (64-bit)	Windows 11 23H2 (64-bit)
CPU	Intel Core i5-8th Gen / AMD Ryzen 5 3000	Intel Core i7-12th Gen / AMD Ryzen 7 5000
RAM	8 GB DDR4	16 GB DDR4 / 32 GB DDR5
GPU	DirectX 12 compatible, 2 GB VRAM	NVIDIA RTX 3060 / AMD RX 6600 (8 GB VRAM)
Storage	10 GB NVMe SSD	50 GB NVMe SSD
Microphone	USB microphone, 16kHz capable	USB condenser microphone, 48kHz
Python	Python 3.10+	Python 3.11.x
PowerShell	PowerShell 5.1	PowerShell 7.4+

Table B.1 — PIVA system requirements

B.2 Core Configuration File

```
# piva_config.yaml — Core Configuration Reference

audio:
  sample_rate: 16000
  channels: 1
  bit_depth: 16
  vad_threshold: 0.5
  vad_min_silence_ms: 600
  noise_suppression: true

voice_cloning:
```



```
  speaker_encoder_model: models/ecapa_tdnn_meta.onnx
  tts_decoder_model: models/piva_vits_decoder.onnx
  vocoder_model: models/hifigan_v2_conditioned.onnx
  embedding_dim: 512
  min_enrollment_utterances: 5
  inference_device: directml  # or cpu

nlu:
  intent_model: models/distilbert_intent.onnx
  entity_model: models/distilbert_ner.onnx
  context_window: 10
  max_sequence_length: 128
  confidence_threshold: 0.75

os_control:
  default_permission_tier: standard
  powershell_timeout_s: 30
  uia_cache_enabled: true
  audit_log_retention_days: 30
  audit_log_encryption: dpapi

security:
  anti_spoof_enabled: true
  anti_spoof_threshold: 0.85
  windows_hello_required_for: [elevated, administrator,
restricted]
  enrollment_data_location: local  # never 'cloud'
```

Code B.1 — PIVA core configuration file reference

B.3 Required Python Dependencies

```
# requirements.txt — PIVA Core Dependencies
onnxruntime-directml>=1.17.0
transformers>=4.38.0
torch>=2.2.0
torchaudio>=2.2.0
pywin32>=306
comtypes>=1.3.0
pyaudio>=0.2.14
silero-vad>=4.0.0
numpy>=1.26.0
scipy>=1.12.0
```



```
pyyaml>=6.0.1
cryptography>=42.0.0
openai-whisper>=20231117
huggingface-hub>=0.21.0
```

Code B.2 — PIVA Python dependency specification

Appendix C — Sample Command Ontology

The following tables provide a representative sample of PIVA's command ontology across the six operational domains. Each entry specifies the canonical command template, required and optional slot types, execution backend, permission tier, and example natural language phrasings.

C.1 File System Commands (Sample)

Command	Required Slots	Optional Slots	Backend	Permission
MoveFile	source_path, dest_path	rename_to	Win32 + PS	Standard
CopyFile	source_path, dest_path	rename_to, overwrite	Win32	Standard
DeleteFile	target_path	permanent (vs recycle)	Win32	Standard
RenameFile	target_path, new_name	—	Win32	Standard
CreateFolder	parent_path, folder_name	—	Win32	Standard
SearchFiles	search_query	location, file_type, date_range	PowerShell	Standard
CompressFiles	source_paths, archive_name	format, password	PowerShell	Standard
ChangePermissions	target_path, permission_rule	recursive	Win32 + PS	Elevated

Table C.1 — Sample File System command ontology entries

C.2 Application Control Commands (Sample)

Command	Required Slots	Optional Slots	Backend	Permission
LaunchApplication	app_name	arguments, window_state	Win32	Standard
CloseApplication	app_name	force_close	Win32	Standard
FocusWindow	app_name	window_title	Win32 UIA	Standard
ResizeWindow	app_name, size_spec	position	Win32	Standard
TileWindows	app_names, layout	—	Win32	Standard
InstallPackage	package_name	version, source	PowerShell	Elevated
UninstallPackage	package_name	—	PowerShell	Elevated
SetStartup	app_name, enabled	—	Registry/PS	Administrator

Table C.2 — Sample Application Control command ontology entries

C.3 System Settings Commands (Sample)

Command	Required Slots	Optional Slots	Backend	Permission
SetVolume	volume_level	device	Win32 API	Standard
SetBrightness	brightness_level	monitor	Win32 API	Standard
SetTheme	theme_name	—	PowerShell	Standard
SetPowerPlan	plan_name	—	PowerShell	Elevated
EnableFeature	feature_name	—	PowerShell	Elevated
ConfigureFirewall	rule_action, rule_target	protocol, port	PowerShell	Administrator
ManageService	service_name, action	startup_type	PowerShell	Administrator
SetNetworkConfig	adapter, setting, value	—	PowerShell	Restricted

Table C.3 — Sample System Settings command ontology entries

References and Bibliography

Page 1 of 3 — Speech Synthesis and Voice Cloning

A. Speech Synthesis Foundations

[1] Oord, A. van den, et al. (2016). WaveNet: A Generative Model for Raw Audio. arXiv preprint arXiv:1609.03499. DeepMind Technical Report. Retrieved from https://arxiv.org/abs/1609.03499

[2] Wang, Y., et al. (2017). Tacotron: Towards End-to-End Speech Synthesis. Proceedings of Interspeech 2017, 4006–4010. doi:10.21437/Interspeech.2017-1452

[3] Shen, J., et al. (2018). Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions. Proceedings of ICASSP 2018, 4779–4783. IEEE. doi:10.1109/ICASSP.2018.8461368

[4] Ren, Y., et al. (2019). FastSpeech: Fast, Robust and Controllable Text to Speech. Advances in Neural Information Processing Systems (NeurIPS) 32, 3171–3180.

[5] Ren, Y., et al. (2021). FastSpeech 2: Fast and High-Quality End-to-End Text-to-Speech. International Conference on Learning Representations (ICLR 2021).

[6] Kim, J., et al. (2021). VITS: Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech. Proceedings of ICML 2021, PMLR 139, 5530–5540.

[7] Kong, J., Kim, J., & Bae, J. (2020). HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. Advances in Neural Information Processing Systems 33.

[8] Casanova, E., et al. (2022). YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for Everyone. Proceedings of ICML 2022, PMLR 162, 2709–2720.

[9] Betker, J. (2023). Better speech synthesis through scaling. arXiv preprint arXiv:2305.07243.

[10] Le, M., et al. (2023). Voicebox: Text-Guided Multilingual Universal Speech Generation at Scale. Advances in Neural Information Processing Systems 36. Meta AI Research.

[11] Wang, C., et al. (2023). Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers (VALL-E). arXiv preprint arXiv:2301.02111. Microsoft Research.

[12] Kharitonov, E., et al. (2023). Speak, Read and Prompt: High-Fidelity Text-to-Speech with Minimal Supervision. Transactions of the Association for Computational Linguistics, 11, 1703–1718.

B. Few-Shot and Meta-Learning for TTS

[13] Chen, S., et al. (2021). AdaSpeech: Adaptive Text to Speech for Custom Voice. International Conference on Learning Representations (ICLR 2021).

[14] Huang, W.-C., et al. (2022). Meta-TTS: Meta-Learning for Few-Shot Speaker Adaptive Text-to-Speech. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 30, 1558–1571.

[15] Finn, C., Abbeel, P., & Levine, S. (2017). Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. Proceedings of ICML 2017, PMLR 70, 1126–1135.

[16] Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical Networks for Few-shot Learning. Advances in Neural Information Processing Systems 30, 4077–4087.

C. Speaker Verification and Identity Encoding

[17] Wan, L., et al. (2018). Generalized End-to-End Loss for Speaker Verification. Proceedings of ICASSP 2018, 4879–4883. IEEE.

[18] Snyder, D., et al. (2018). X-vectors: Robust DNN Embeddings for Speaker Recognition. Proceedings of ICASSP 2018, 5329–5333. IEEE.

[19] Desplanques, B., Thienpondt, J., & Demuynck, K. (2020). ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification. Proceedings of Interspeech 2020, 3830–3834.

[20] Nagrani, A., Chung, J. S., & Zisserman, A. (2017). VoxCeleb: A Large-Scale Speaker Identification Dataset. Proceedings of Interspeech 2017, 2616–2620.

[21] Chung, J. S., et al. (2018). VoxCeleb2: Deep Speaker Recognition. Proceedings of Interspeech 2018, 1086–1090.

D. Natural Language Understanding

[22] Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of NAACL-HLT 2019, 4171–4186. Association for Computational Linguistics.

[23] Sanh, V., et al. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108. Hugging Face.

[24] Coucke, A., et al. (2018). Snips Voice Platform: an embedded Spoken Language Understanding system for private-by-design voice interfaces. arXiv preprint arXiv:1805.10190.

[25] Hemphill, C. T., Godfrey, J. J., & Doddington, G. R. (1990). The ATIS spoken language systems pilot corpus. Proceedings of the Workshop on Speech and Natural Language, 96–101. ACL.

[26] Liu, B., & Lane, I. (2016). Attention-Based Recurrent Neural Network Models for Joint Intent Detection and Slot Filling. Proceedings of Interspeech 2016, 685–689.

[27] Chen, Q., et al. (2019). BERT for Joint Intent Classification and Slot Filling. arXiv preprint arXiv:1902.10909.

[28] Mrkšić, N., et al. (2017). Neural Belief Tracker: Data-Driven Dialogue State Tracking. Proceedings of ACL 2017, 1777–1788. Association for Computational Linguistics.

[29] Lafferty, J., McCallum, A., & Pereira, F. (2001). Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data. Proceedings of ICML 2001, 282–289.

E. Automatic Speech Recognition

[30] Radford, A., et al. (2023). Robust Speech Recognition via Large-Scale Weak Supervision. Proceedings of ICML 2023, PMLR 202, 28492–28518. OpenAI.

[31] Baevski, A., et al. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. Advances in Neural Information Processing Systems 33. Facebook AI Research.

[32] Zhang, Y., et al. (2022). Benchmarking Large Language Models in Complex Instruction-Following. arXiv preprint arXiv:2307.15337.

F. Voice-Controlled Computing and OS Automation

[33] Shneiderman, B. (2000). The limits of speech recognition. Communications of the ACM, 43(9), 63–65. doi:10.1145/348941.348990

[34] Microsoft Corporation. (2023). UI Automation Overview. Microsoft Documentation. Retrieved from https://learn.microsoft.com/en-us/dotnet/framework/ui-automation/ui-automation-overview

[35] Microsoft Corporation. (2024). Windows Audio Session API (WASAPI). Microsoft Documentation. Retrieved from https://learn.microsoft.com/en-us/windows/win32/coreaudio/wasapi

[36] Microsoft Corporation. (2024). PowerShell 7 Documentation. GitHub. Retrieved from https://github.com/PowerShell/PowerShell/tree/master/docs

[37] Bolt, R. A. (1980). Put-That-There: Voice and gesture at the graphics interface. ACM SIGGRAPH Computer Graphics, 14(3), 262–270.

[38] Myers, B. A., et al. (2004). Extending the Windows API with new Input Devices and Accessibility Features. CHI 2004 Extended Abstracts, 1023–1024.

[39] Bangor, A., Kortum, P., & Miller, J. (2009). Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale. Journal of Usability Studies, 4(3), 114–123.

G. Privacy, Security, and Ethics

[40] Nautsch, A., et al. (2019). Preserving Privacy with Generative Adversarial Networks for Speaker De-Identification. IEEE Signal Processing Letters, 27, 141–145.

[41] Müller, N. M., et al. (2022). Does Audio Deepfake Detection Generalize? Proceedings of Interspeech 2022, 2783–2787.

[42] Yi, J., et al. (2022). ADD 2022: The First Audio Deep Synthesis Detection Challenge. arXiv preprint arXiv:2202.08433.

[43] European Parliament and Council. (2024). Regulation (EU) 2024/1689 — Artificial Intelligence Act. Official Journal of the European Union. Retrieved from https://eur-lex.europa.eu/

[44] California Legislature. (2024). AB 2602 — Contracts with digital replicas: minors. California Legislative Information. Retrieved from https://leginfo.legislature.ca.gov/

[45] European Parliament and Council. (2016). Regulation (EU) 2016/679 — General Data Protection Regulation (GDPR). Official Journal of the European Union.

[46] Stokel-Walker, C. (2023). The Sticky Ethics of Synthetic Voices. Nature, 623, 246–248. doi:10.1038/d41586-023-03425-4

H. Machine Learning Frameworks and Tooling

[47] Bai, J., et al. (2019). ONNX: Open Neural Network Exchange. GitHub. Retrieved from https://github.com/onnx/onnx

[48] Microsoft Corporation. (2023). ONNX Runtime Documentation. GitHub. Retrieved from https://onnxruntime.ai/docs/

[49] Paszke, A., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems 32, 8026–8037.

[50] Wolf, T., et al. (2020). Transformers: State-of-the-Art Natural Language Processing. Proceedings of EMNLP 2020 System Demonstrations, 38–45. Hugging Face.

[51] Loshchilov, I., & Hutter, F. (2019). Decoupled Weight Decay Regularization. International Conference on Learning Representations (ICLR 2019).

[52] Kim, S., & Smaragdis, P. (2022). Neural Vocoder Is All You Need for Speech Super-Resolution. arXiv preprint arXiv:2203.14941.

I. Datasets and Benchmarks

[53] Panayotov, V., et al. (2015). Librispeech: An ASR Corpus Based on Public Domain Audio Books. Proceedings of ICASSP 2015, 5206–5210. IEEE.

[54] Veaux, C., Yamagishi, J., & MacDonald, K. (2017). CSTR VCTK Corpus: English Multi-Speaker Corpus for CSTR Voice Cloning Toolkit. University of Edinburgh. The Centre for Speech Technology Research (CSTR).

[55] Bakhturina, E., et al. (2021). Hi-Fi Multi-Speaker English TTS Dataset. Proceedings of Interspeech 2021, 2776–2780.

[56] Richey, C., et al. (2018). Voices Obscured in Complex Environmental Settings (VOICES) corpus. arXiv preprint arXiv:1811.06681.

[57] Thiemann, J., Ito, N., & Vincent, E. (2013). The diverse environments multi-channel acoustic noise database (DEMAND). Proceedings of Meetings on Acoustics, 19(1), 035081.

J. Human-Computer Interaction

[58] Reeves, B., & Nass, C. (1996). The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places. Cambridge University Press.

[59] Nass, C., & Brave, S. (2005). Wired for Speech: How Voice Activates and Advances the Human-Computer Relationship. MIT Press.

[60] Porcheron, M., et al. (2018). Voice Interfaces in Everyday Life. Proceedings of CHI 2018. ACM. doi:10.1145/3173574.3174214

[61] Luger, E., & Sellen, A. (2016). Like Having a Really Bad PA: The Gulf between User Expectation and Experience of Conversational Agents. Proceedings of CHI 2016, 5286–5297. ACM.

[62] Lopatovska, I., et al. (2019). Talk to me: Exploring user interactions with the Amazon Alexa. Journal of Librarianship and Information Science, 51(4), 984–997.

K. Related Frameworks and Patents

[63] Amodei, D., et al. (2016). Deep Speech 2: End-to-End Speech Recognition in English and Mandarin. Proceedings of ICML 2016, PMLR 48, 173–182. Baidu Research.

[64] Google Inc. (2021). Duplex: AI System for Natural Conversations. Google AI Blog. Retrieved from https://ai.google/research/pubs/

[65] Apple Inc. (2023). Siri Natural Language Framework Documentation. Apple Developer Documentation. Retrieved from https://developer.apple.com/documentation/

[66] Amazon Web Services. (2024). Amazon Alexa Skills Kit Documentation. AWS Documentation. Retrieved from https://developer.amazon.com/en-US/alexa/

[67] ITU-T. (2001). Recommendation P.862: Perceptual evaluation of speech quality (PESQ). International Telecommunication Union.

[68] Taal, C. H., et al. (2011). An Algorithm for Intelligibility Prediction of Time-Frequency Weighted Noisy Speech. IEEE Transactions on Audio, Speech, and Language Processing, 19(7), 2125–2136.

[69] Loizou, P. C. (2013). Speech Enhancement: Theory and Practice (2nd ed.). CRC Press. ISBN: 978-1-4665-0421-9

[70] Schuller, B. W., et al. (2021). Speech and Language Technology for Mental Health and Psychotherapy. IEEE Signal Processing Magazine, 38(6), 56–66.

— End of References —

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